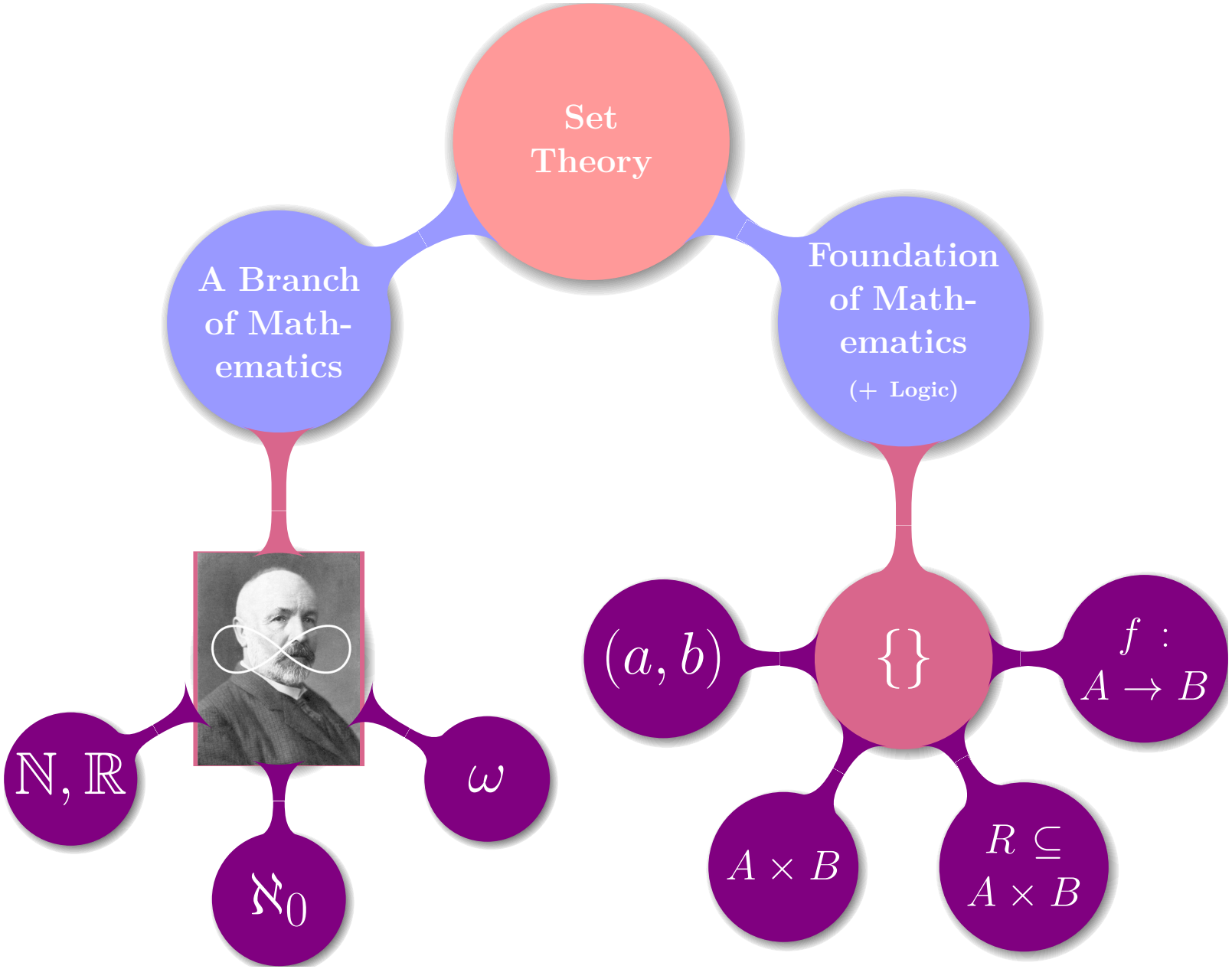
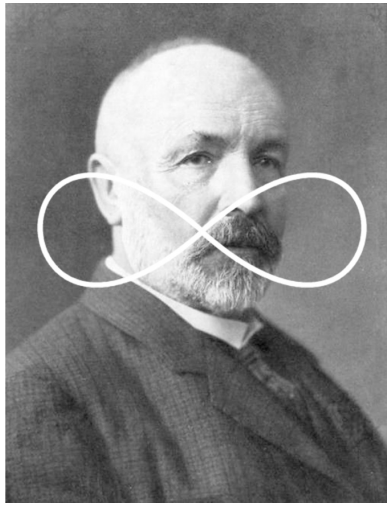
6-infinity: 无穷 (Infinity)

魏恒峰

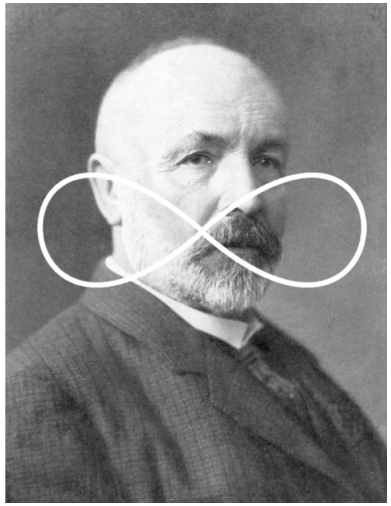
hfwei@nju.edu.cn

2026 年 04 月 14 日


Set Theory
A Branch of Mathematics
Foundation of Mathematics
(+ Logic)
$\mathbb{N}, \mathbb{R}$
$\aleph_0$
ω
(a, b)
$\{\}$
$f : A \to B$
$A \times B$
$R \subseteq A \times B$

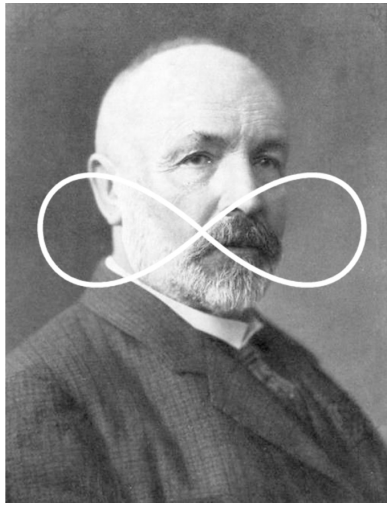


Georg Cantor (1845 – 1918)



Georg Cantor (1845 – 1918)

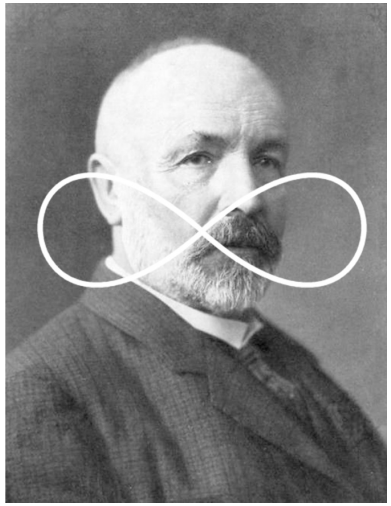
Leopold Kronecker
(1823 – 1891)



Georg Cantor (1845 – 1918)

Leopold Kronecker
(1823 – 1891)

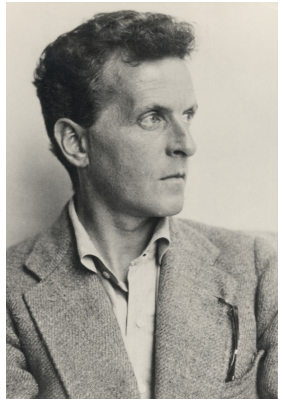
Henri Poincaré
(1854 – 1912)



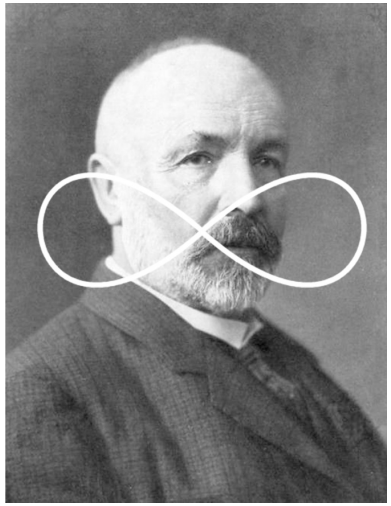
Georg Cantor (1845 – 1918)

Leopold Kronecker
(1823 – 1891)

Henri Poincaré
(1854 – 1912)



Ludwig Wittgenstein
(1889 – 1951)

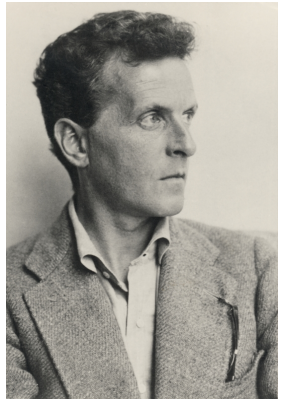


Georg Cantor (1845 – 1918)

David Hilbert (1862 – 1943)

Leopold Kronecker
(1823 – 1891)

Henri Poincaré
(1854 – 1912)



Ludwig Wittgenstein
(1889 – 1951)

From his paradise that Cantor with us unfolded, we hold our breath in awe; knowing, we shall not be expelled.

— *David Hilbert*

"没有人能把我们从 *Cantor* 创造的乐园中驱逐出去"

From his paradise that Cantor with us unfolded, we hold our breath in awe; knowing, we shall not be expelled.

— *David Hilbert*

“没有人能把我们从 *Cantor* 创造的乐园中驱逐出去”

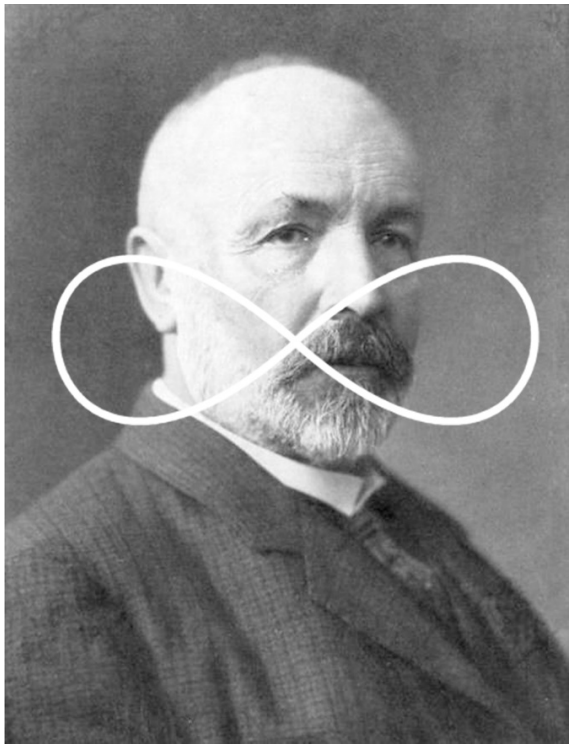
GEORG CANTOR
MATHEMATIKER
BEGRÜNDER DER MENGENLEHRE
1845-1918
DAS WESEN DER MATHEMA
TIK LIEGT IN IHRER FREIHEIT
GEORG CANTOR

"das wesen der mathematik liegt in ihrer freiheit"

"das wesen der mathematik liegt in ihrer freiheit"

"The essence of mathematics lies in its freedom"

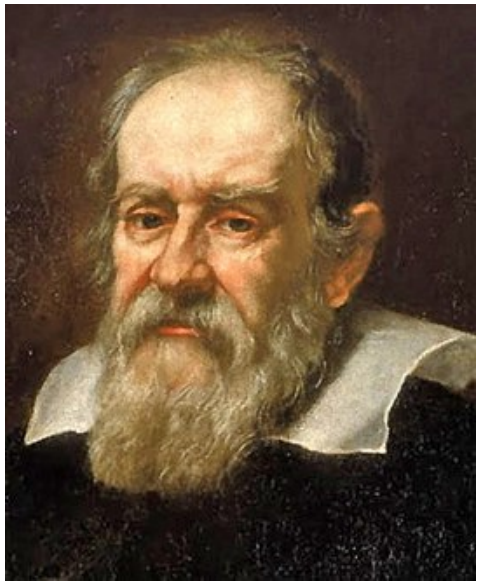
Before Cantor



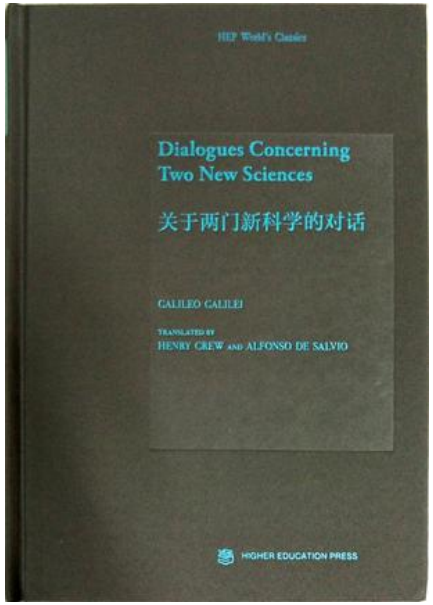
一部上帝安排我们现有空间秩序的方案之书
决定经典
001
一部高度展示人类的逻辑理性与逻辑思维能力的体系教本
全新修订版
[古希腊] 欧几里得 / 著
几何原本
建立空间秩序最久远最权威的逻辑推演语系
Euclid's Elements
13卷图释全本
燕晓东 / 译
凤凰出版传媒集团
江苏人民出版社
凤凰联动 FONGHONG
凤凰决定 DECISION

一部上帝安排我们现有空间秩序的方案之书
决定经典
001
一部高度展示人类的逻辑理性与逻辑思维能力的体系教本
全新修订版
[古希腊] 欧几里得 / 著
几何原本
建立空间秩序最久远最权威的逻辑推演语系
Euclid's Elements
13卷图释全本
燕晓东 / 译
凤凰出版传媒集团
江苏人民出版社
凤凰联动 FONGHONG
凤凰决定 DECISION

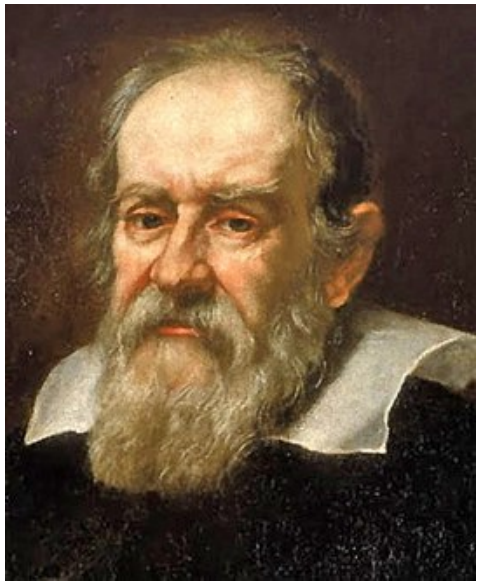
公理: “整体大于部分”



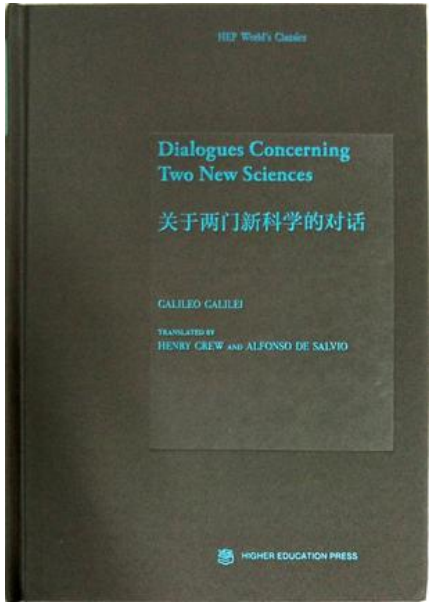
Galileo Galilei (1564 – 1642)



“关于两门新科学的对话” (1638)



Galileo Galilei (1564 – 1642)



“关于两门新科学的对话” (1638)

“用我们有限的心智来讨论无限 · · · ”

$$S_1 = \{1, 2, 3, \cdots, n, \cdots\}$$

$$S_2 = \{1, 4, 9, \cdots, n^2, \cdots\}$$

$$S_1 = \{1, 2, 3, \cdots, n, \cdots\}$$
$$S_2 = \{1, 4, 9, \cdots, n^2, \cdots\}$$

$|S_1| = |S_2|$ $\qquad$ $S_2 \subset S_1$

$$S_1 = \{1, 2, 3, \cdots, n, \cdots\}$$
$$S_2 = \{1, 4, 9, \cdots, n^2, \cdots\}$$

$|S_1| = |S_2|$ $\quad$ $S_2 \subset S_1$

“部分等于全体”

$$S_1 = \{1, 2, 3, \cdots, n, \cdots\}$$

$$S_2 = \{1, 4, 9, \cdots, n^2, \cdots\}$$

$|S_1| = |S_2|$ $\quad$ $S_2 \subset S_1$

"部分等于全体"

吓得我吃了一鲸

$$S_1 = \{1, 2, 3, \cdots, n, \cdots\}$$
$$S_2 = \{1, 4, 9, \cdots, n^2, \cdots\}$$

$|S_1| = |S_2|$ $\quad$ $S_2 \subset S_1$

"部分等于全体"

吓得我吃了一鲸

说到底，"等于"、"大于"和"小于"诸性质不能用于无限，而只能用于有限的数量。 — *Galileo Galilei*

$$S_1 = \{1, 2, 3, \cdots, n, \cdots\}$$
$$S_2 = \{1, 4, 9, \cdots, n^2, \cdots\}$$

$|S_1| = |S_2|$ $\quad$ $S_2 \subset S_1$

"部分等于全体"

吓得我吃了一鲸

说到底，"等于"、"大于"和"小于"诸性质不能用于无限，而只能用于有限的数量。 — *Galileo Galilei*

无穷数是不可能的。 — *Gottfried Wilhelm Leibniz*

这些证明一开始就期望那些数要具有有穷数的一切性质，或者甚至于把有穷数的性质强加于无穷。

这些证明一开始就期望那些数要具有有穷数的一切性质，或者甚至于把有穷数的性质强加于无穷。

相反，这些无穷数，如果它们能够以任何形式被理解的话，倒是由于它们与有穷数的对应，它们必须具有完全新的数量特征。

这些证明一开始就期望那些数要具有有穷数的一切性质，或者甚至于把有穷数的性质强加于无穷。

相反，这些无穷数，如果它们能够以任何形式被理解的话，倒是由于它们与有穷数的对应，它们必须具有完全新的数量特征。

这些性质完全依赖于事物的本性，⋯ 而并非来自我们的主观任意性或我们的偏见。

— *Georg Cantor (1885)*

Definition (Dedekind-infinite & Dedekind-finite (Dedekind, 1888))

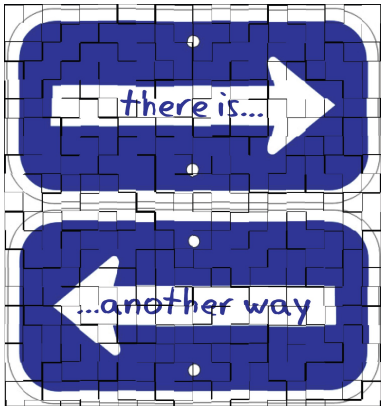
A set A is *Dedekind-infinite* if there is a bijective function from A to some proper subset B of A.

A set is *Dedekind-finite* if it is not Dedekind-infinite.

Definition (Dedekind-infinite & Dedekind-finite (Dedekind, 1888))

A set A is *Dedekind-infinite* if there is a bijective function from A to some proper subset B of A.

A set is *Dedekind-finite* if it is not Dedekind-infinite.



This is a theorem in our theory of infinity.

We have not defined “finite” and “infinite”!

Comparing Sets

Comparing Sets

$$=$$

$$<$$

Comparing Sets

$=$ $<$

Function

[redacted]

[redacted]

Definition ($|A| = |B|$ ($A \approx B$) (1878))

A and B are *equipotent (等势)* if there exists a *bijection* from A to B.

Definition ($|A| = |B|$ ($A \approx B$) (1878))

A and B are *equipotent (等势)* if there exists a *bijection* from A to B.

$$\overline{\overline{A}} \quad \text{(two abstractions)}$$

Definition ($|A| = |B|$ ($A \approx B$) (1878))

A and B are *equipotent (等势)* if there exists a *bijection* from A to B.

$$\overline{\overline{A}} \quad \text{(two abstractions)}$$

Abstract from elements: $\{1, 2, 3\}$ *vs.* $\{a, b, c\}$

Definition ($|A| = |B|$ ($A \approx B$) (1878))

A and B are *equipotent (等势)* if there exists a *bijection* from A to B.

$$\overline{\overline{A}} \quad \text{(two abstractions)}$$

Abstract from elements: $\{1, 2, 3\} \quad vs. \quad \{a, b, c\}$

Abstract from order: $\{1, 2, 3, \cdots\} \quad vs. \quad \{1, 3, 5, \cdots, 2, 4, 6, \cdots\}$

Definition ($|A| = |B|$ ($A \approx B$) (1878))

A and B are *equipotent* if there exists a *bijection* from A to B.

Q : Is "$\approx$" an equivalence relation?

Definition ($|A| = |B|$ ($A \approx B$) (1878))

A and B are *equipotent* if there exists a *bijection* from A to B.

Q : Is "$\approx$" an equivalence relation?

Theorem ( )

For any sets A, B, C:

(a) $A \approx B$

(b) $A \approx B \implies B \approx A$

(c) $A \approx B \wedge B \approx C \implies A \approx C$

Definition ($|A| = |B|$ ($A \approx B$) (1878))

A and B are *equipotent* if there exists a *bijection* from A to B.

Q : Is "$\approx$" an equivalence relation?

Theorem (The "Equivalence Concept" of Equipotent)

For any sets A, B, C:

(a) $A \approx B$

(b) $A \approx B \implies B \approx A$

(c) $A \approx B \wedge B \approx C \implies A \approx C$

Definition ($|A| = |B|$ ($A \approx B$) (1878))

A and B are *equipotent* if there exists a *bijection* from A to B.

Q : Is "$\approx$" an equivalence relation?

Theorem (The "Equivalence Concept" of Equipotent)

For any sets A, B, C:

(a) $A \approx B$

(b) $A \approx B \implies B \approx A$

(c) $A \approx B \wedge B \approx C \implies A \approx C$

Do not care too much.

Definition (Finite)

X is finite if

$$\exists n \in \mathbb{N} : |X| = |n| = |\{0, 1, \ldots, n-1\}|.$$

Definition (Finite)

X is finite if

$$\exists n \in \mathbb{N} : |X| = |n| = |\{0, 1, \ldots, n-1\}|.$$

简记为 $|X| = n$

Definition (Finite)

X is finite if

$$\exists n \in \mathbb{N} : |X| = |n| = |\{0, 1, \ldots, n-1\}|.$$

简记为 $|X| = n$

集合 X 是有穷的当且仅当它与某个自然数等势。

Definition (Infinite)

X is infinite if it is not finite:

$$\forall n \in \mathbb{N} : |X| \neq n.$$

Definition (Infinite)

X is infinite if it is not finite:

$$\forall n \in \mathbb{N} : |X| \neq n.$$

Theorem (Existence of Infinite Sets!)

$\mathbb{N}$ *is infinite.*

Definition (Infinite)

X is infinite if it is not finite:

$$\forall n \in \mathbb{N} : |X| \neq n.$$

Theorem (Existence of Infinite Sets!)

$\mathbb{N}$ *is infinite.*

By Contradiction.

Definition (Infinite)

X is infinite if it is not finite:

$$\forall n \in \mathbb{N} : |X| \neq n.$$

Theorem (Existence of Infinite Sets!)

$\mathbb{N}$ *is infinite.*

By Contradiction.

$$\exists n \in \mathbb{N} : |\mathbb{N}| = n.$$

Definition (Infinite)

X is infinite if it is not finite:

$$\forall n \in \mathbb{N} : |X| \neq n.$$

Theorem (Existence of Infinite Sets!)

$\mathbb{N}$ *is infinite.*

By Contradiction.

$$\exists n \in \mathbb{N} : |\mathbb{N}| = n.$$

$$\exists f : \mathbb{N} \xleftrightarrow[onto]{1-1} \{0, 1, \cdots, n-1\}$$

Definition (Infinite)

X is infinite if it is not finite:

$$\forall n \in \mathbb{N} : |X| \neq n.$$

Theorem (Existence of Infinite Sets!)

$\mathbb{N}$ *is infinite.*

By Contradiction.

$$\exists n \in \mathbb{N} : |\mathbb{N}| = n.$$

$$\exists f : \mathbb{N} \xleftrightarrow[onto]{1-1} \{0, 1, \cdots, n-1\}$$

$$g \triangleq f|_{\{0,1,\ldots,n\}} : \{0, 1, \cdots, n\} \to \{0, 1, \cdots, n-1\}$$

Definition (Infinite)

X is infinite if it is not finite:

$$\forall n \in \mathbb{N} : |X| \neq n.$$

Theorem (Existence of Infinite Sets!)

$\mathbb{N}$ *is infinite.*

By Contradiction.

$$\exists n \in \mathbb{N} : |\mathbb{N}| = n.$$

$$\exists f : \mathbb{N} \xleftrightarrow[onto]{1-1} \{0, 1, \cdots, n-1\}$$

$$g \triangleq f|_{\{0,1,\ldots,n\}} : \{0, 1, \cdots, n\} \to \{0, 1, \cdots, n-1\}$$

By the Pigeonhole Principle : g is not 1-1

Definition (Infinite)

X is infinite if it is not finite:

$$\forall n \in \mathbb{N} : |X| \neq n.$$

Theorem (Existence of Infinite Sets!)

$\mathbb{N}$ *is infinite.*

By Contradiction.

$$\exists n \in \mathbb{N} : |\mathbb{N}| = n.$$

$$\exists f : \mathbb{N} \xleftrightarrow[onto]{1-1} \{0, 1, \cdots, n-1\}$$

$$g \triangleq f|_{\{0,1,\ldots,n\}} : \{0, 1, \cdots, n\} \to \{0, 1, \cdots, n-1\}$$

By the Pigeonhole Principle : g is not 1-1 $\implies$ f is not 1-1

Definition (Infinite)

X is infinite if it is not finite:

$$\forall n \in \mathbb{N} : |X| \neq n.$$

Theorem (Existence of Infinite Sets!)

$\mathbb{N}$ *is infinite.* *(So are $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$.)*

By Contradiction.

$$\exists n \in \mathbb{N} : |\mathbb{N}| = n.$$

$$\exists f : \mathbb{N} \xleftrightarrow[onto]{1-1} \{0, 1, \cdots, n-1\}$$

$$g \triangleq f|_{\{0,1,\ldots,n\}} : \{0, 1, \cdots, n\} \to \{0, 1, \cdots, n-1\}$$

By the Pigeonhole Principle : g is not 1-1 $\implies$ f is not 1-1

Definition (Infinite)

For any set X,

Countably Infinite

$$|X| = |\mathbb{N}| \triangleq \aleph_0$$

Definition (Infinite)

For any set X,

Countably Infinite

$$|X| = |\mathbb{N}| \triangleq \aleph_0$$

Countable

$$(\text{finite} \vee \text{countably infinite})$$

Definition (Infinite)

For any set X,

Countably Infinite

$$|X| = |\mathbb{N}| \triangleq \aleph_0$$

Countable

$$(\text{finite} \vee \text{countably infinite})$$

Uncountable

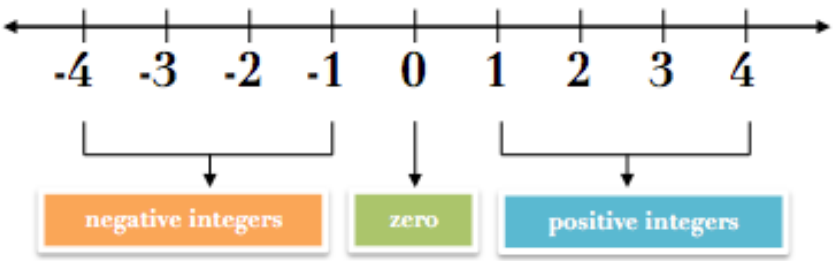
$$(\neg \text{ countable})$$

$$(\text{infinite}) \wedge \Big(\neg\ (\text{countably infinite})\Big)$$

$$\aleph_0$$

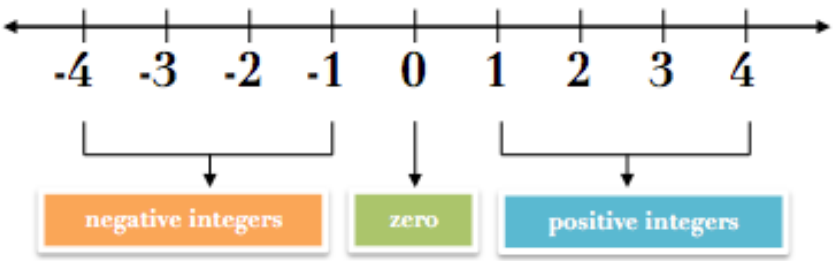
Theorem ($\mathbb{Z}$ is Countable.)

$$|\mathbb{Z}| = |\mathbb{N}|$$



Theorem ($\mathbb{Z}$ is Countable.)

$$|\mathbb{Z}| = |\mathbb{N}|$$



$$0 \quad 1 \quad -1 \quad 2 \quad -2 \quad \cdots$$

Theorem ($\mathbb{Q}$ is Countable. (Cantor 1873-11; Published in 1874))

$$|\mathbb{Q}| = |\mathbb{N}|$$

Theorem ($\mathbb{Q}$ is Countable. (Cantor 1873-11; Published in 1874))

$$|\mathbb{Q}| = |\mathbb{N}|$$

$|\mathbb{Q}| = |\mathbb{N}|$

$$q \in \mathbb{Q}^+ : a/b \ (a, b \in \mathbb{N}^+)$$

Theorem ($\mathbb{Q}$ is Countable. (Cantor 1873-11; Published in 1874))

$$|\mathbb{Q}| = |\mathbb{N}|$$

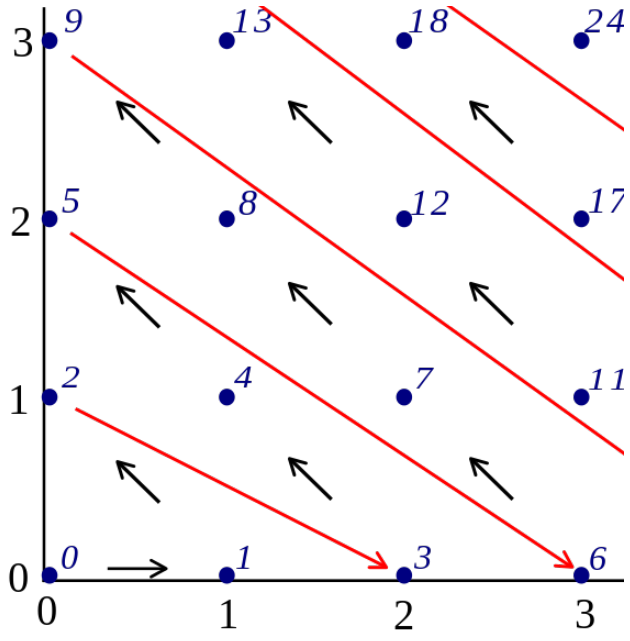
$|\mathbb{Q}| = |\mathbb{N}|$

$$q \in \mathbb{Q}^+ : a/b \ (a, b \in \mathbb{N}^+)$$

$$\begin{array}{ccccccccc}
1/1 & 1/2 & 1/3 & 1/4 & 1/5 & 1/6 & 1/7 & 1/8 & \cdots \\
2/1 & 2/2 & 2/3 & 2/4 & 2/5 & 2/6 & 2/7 & 2/8 & \cdots \\
3/1 & 3/2 & 3/3 & 3/4 & 3/5 & 3/6 & 3/7 & 3/8 & \cdots \\
4/1 & 4/2 & 4/3 & 4/4 & 4/5 & 4/6 & 4/7 & 4/8 & \cdots \\
5/1 & 5/2 & 5/3 & 5/4 & 5/5 & 5/6 & 5/7 & 5/8 & \cdots \\
6/1 & 6/2 & 6/3 & 6/4 & 6/5 & 6/6 & 6/7 & 6/8 & \cdots \\
7/1 & 7/2 & 7/3 & 7/4 & 7/5 & 7/6 & 7/7 & 7/8 & \cdots \\
8/1 & 8/2 & 8/3 & 8/4 & 8/5 & 8/6 & 8/7 & 8/8 & \cdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots
\end{array}$$

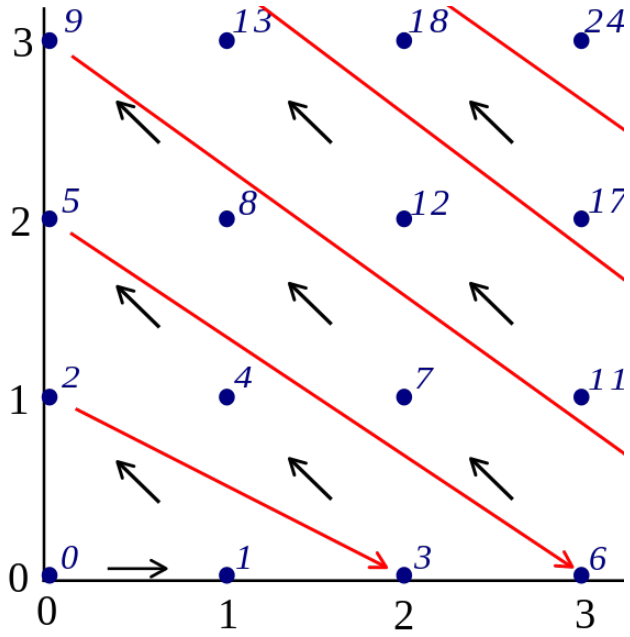
Theorem ($\mathbb{N} \times \mathbb{N}$ is Countable.)

$$|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$$



Theorem ($\mathbb{N} \times \mathbb{N}$ is Countable.)

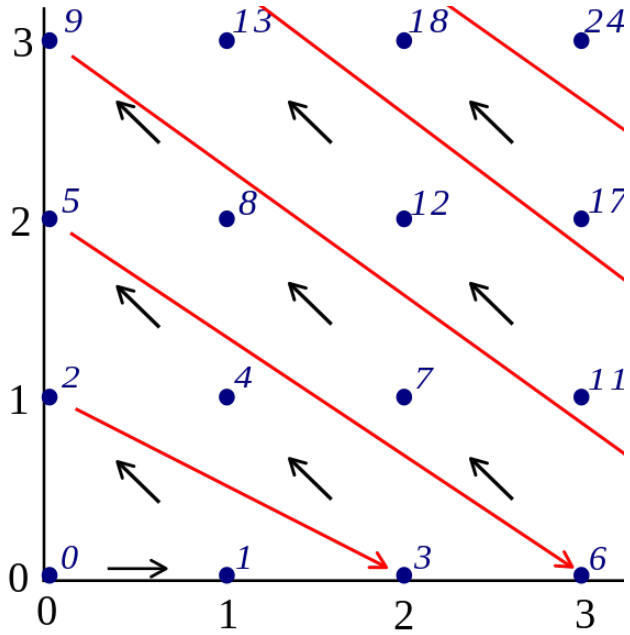
$$|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$$



$$\pi : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$$

Theorem ($\mathbb{N} \times \mathbb{N}$ is Countable.)

$$|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$$



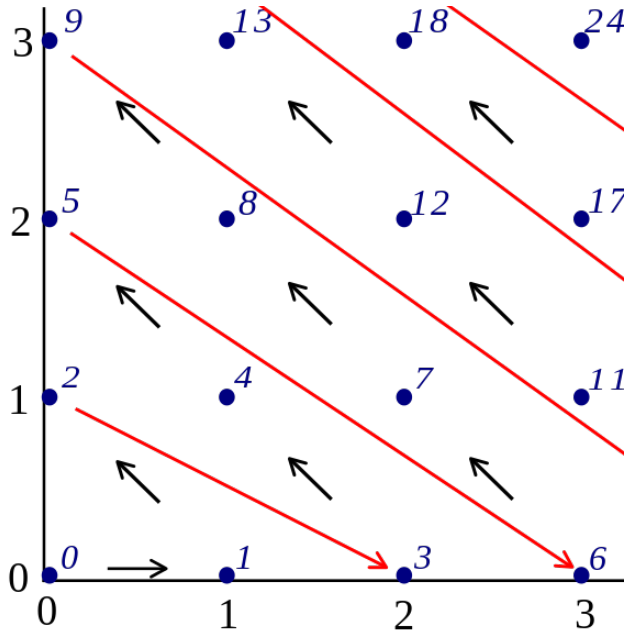
$$\pi : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$$

$$\pi(k_1, k_2) = \frac{1}{2}(k_1+k_2)(k_1+k_2+1)+k_2$$

$$\pi(2, 1) = 7$$

Theorem ($\mathbb{N} \times \mathbb{N}$ is Countable.)

$$|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$$



$$\pi : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$$

$$\pi(k_1, k_2) = \frac{1}{2}(k_1+k_2)(k_1+k_2+1)+k_2$$

$$\pi(2, 1) = 7$$

Cantor Pairing Function

Theorem ($\mathbb{N}^n$ is Countable.)

$$|\mathbb{N}^n| = |\mathbb{N}|$$

Theorem ($\mathbb{N}^n$ is Countable.)

$$|\mathbb{N}^n| = |\mathbb{N}|$$

Theorem

The Cartesian product of finitely many countable sets is countable.

Theorem ($\mathbb{N}^n$ is Countable.)

$$|\mathbb{N}^n| = |\mathbb{N}|$$

Theorem

The Cartesian product of finitely many countable sets is countable.

$$\mathbb{N}^n \quad vs. \quad \mathbb{N}^{\mathbb{N}}$$

Theorem ($\mathbb{N}^n$ is Countable.)

$$|\mathbb{N}^n| = |\mathbb{N}|$$

Theorem

The Cartesian product of finitely many countable sets is countable.

$$\mathbb{N}^n \quad vs. \quad \mathbb{N}^{\mathbb{N}}$$

$$\pi^{(n)} : \mathbb{N}^n \to \mathbb{N}$$

Theorem ($\mathbb{N}^n$ is Countable.)

$$|\mathbb{N}^n| = |\mathbb{N}|$$

Theorem

The Cartesian product of finitely many countable sets is countable.

$$\mathbb{N}^n \quad vs. \quad \mathbb{N}^{\mathbb{N}}$$

$$\pi^{(n)} : \mathbb{N}^n \to \mathbb{N}$$

$$\pi^2(k_1, k_2) = \frac{1}{2}(k_1 + k_2)(k_1 + k_2 + 1) + k_2$$

Theorem ($\mathbb{N}^n$ is Countable.)

$$|\mathbb{N}^n| = |\mathbb{N}|$$

Theorem

The Cartesian product of finitely many countable sets is countable.

$$\mathbb{N}^n \quad vs. \quad \mathbb{N}^{\mathbb{N}}$$

$$\pi^{(n)} : \mathbb{N}^n \to \mathbb{N}$$

$$\pi^2(k_1, k_2) = \frac{1}{2}(k_1 + k_2)(k_1 + k_2 + 1) + k_2$$

$$\pi^{(n)}(k_1, \ldots, k_{n-1}, k_n) = \pi(\pi^{(n-1)}(k_1, \ldots, k_{n-1}), k_n) \quad (n \geq 3)$$

Theorem

Any finite union of countable sets is countable.

Theorem

Any finite union of countable sets is countable.

$$A = \{a_n \mid n \in \mathbb{N}\} \quad B = \{b_n \mid n \in \mathbb{N}\} \quad C = \{c_n \mid n \in \mathbb{N}\}$$

Theorem

Any finite union of countable sets is countable.

$$A = \{a_n \mid n \in \mathbb{N}\} \quad B = \{b_n \mid n \in \mathbb{N}\} \quad C = \{c_n \mid n \in \mathbb{N}\}$$

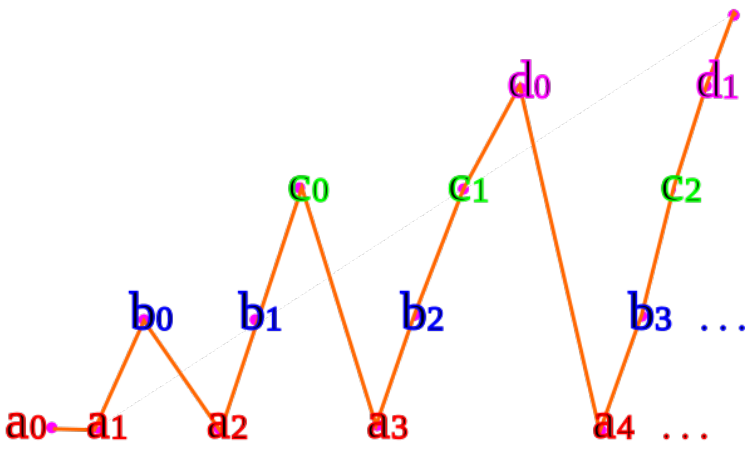
$$a_0 \quad b_0 \quad c_0 \quad a_1 \quad b_1 \quad c_1 \cdots$$

Theorem

The union of countably many countable sets is countable.

Theorem

The union of countably many countable sets is countable.



Counting by Diagonals

Beyond

$$\aleph_0$$

Theorem ($\mathbb{R}$ is Uncountable. (Cantor 1873-12; Published in 1874))

$$|\mathbb{R}| \neq |\mathbb{N}|$$

Theorem ($\mathbb{R}$ is Uncountable. (Cantor 1873-12; Published in 1874))

$$|\mathbb{R}| \neq |\mathbb{N}|$$

Theorem ($\mathbb{R}$ is Uncountable. (Cantor 1873-12; Published in 1874))

$$|\mathbb{R}| \neq |\mathbb{N}|$$

Different "Sizes" of Infinity

Theorem ($\mathbb{R}$ is Uncountable. (Cantor 1873-12; Published in 1874))

$$|\mathbb{R}| \neq |\mathbb{N}|$$

Different "Sizes" of Infinity

Cantor's Diagonal Argument (1890)

Theorem ($\mathbb{R}$ is Uncountable. (Cantor 1873-12; Published in 1874))

$$|\mathbb{R}| \neq |\mathbb{N}|$$

Theorem ($\mathbb{R}$ is Uncountable. (Cantor 1873-12; Published in 1874))

$$|\mathbb{R}| \neq |\mathbb{N}|$$

By Contradiction.

Theorem ($\mathbb{R}$ is Uncountable. (Cantor 1873-12; Published in 1874))

$$|\mathbb{R}| \neq |\mathbb{N}|$$

By Contradiction.

$$f : \mathbb{R} \xleftrightarrow[onto]{1-1} \mathbb{N}$$

Theorem ($\mathbb{R}$ is Uncountable. (Cantor 1873-12; Published in 1874))

$$|\mathbb{R}| \neq |\mathbb{N}|$$

By Contradiction.

$$f : \mathbb{R} \xleftrightarrow[onto]{1-1} \mathbb{N}$$

3.14159...
1.41421...
1.73205...
2.23606...
2.71828...
0.14285...

$\Downarrow$

3.43625...

$\Downarrow$

2.32514...

$$\mathfrak{c} \triangleq |\mathbb{R}|$$

$$\mathfrak{c} \triangleq |\mathbb{R}|$$

$\mathbb{R}$ 是一个连续统 (Continuum)

https://en.wikipedia.org/wiki/Continuum_(set_theory)

Theorem ($|\mathbb{R}|$ (Cantor 1877))

$$|(0,1)| = |\mathbb{R}| = |\mathbb{R} \times \mathbb{R}| = |\mathbb{R}^{n \in \mathbb{N}}|$$

Theorem ($|\mathbb{R}|$ (Cantor 1877))

$$|(0,1)| = |\mathbb{R}| = |\mathbb{R} \times \mathbb{R}| = |\mathbb{R}^{n \in \mathbb{N}}|$$

$$f(x) = \tan \frac{(2x-1)\pi}{2}$$

$$|(0,1)| = |(-\frac{\pi}{2}, \frac{\pi}{2})| = |\mathbb{R}|$$

$$\mathbb{R} \times \mathbb{R} \approx \mathbb{R}$$

$$\mathbb{R} \times \mathbb{R} \approx \mathbb{R}$$

$$(x = 0.a_1a_2a_3\cdots, y = 0.b_1b_2b_3\cdots)$$

$$\mathbb{R} \times \mathbb{R} \approx \mathbb{R}$$

$$(x = 0.a_1a_2a_3\cdots, y = 0.b_1b_2b_3\cdots) \mapsto 0.a_1b_1a_2b_2a_3b_3\cdots$$

$$\mathbb{R} \times \mathbb{R} \approx \mathbb{R}$$

$$(x = 0.a_1a_2a_3\cdots, y = 0.b_1b_2b_3\cdots) \mapsto 0.a_1b_1a_2b_2a_3b_3\cdots$$

Was Cantor Surprised?

Theorem ($|\mathbb{R}|$ (Cantor 1877))

$$|(0,1)| = |\mathbb{R}| = |\mathbb{R} \times \mathbb{R}| = |\mathbb{R}^n|$$

"Je le vois, mais je ne le crois pas !"

"I see it, but I don't believe it !"

— *Cantor's letter to Dedekind (1877).*

Theorem ($|\mathbb{R}|$ (Cantor 1877))

$$|(0,1)| = |\mathbb{R}| = |\mathbb{R} \times \mathbb{R}| = |\mathbb{R}^n|$$

"Je le vois, mais je ne le crois pas !"

"I see it, but I don't believe it !"

— *Cantor's letter to Dedekind (1877).*

Q : Then, what is "dimension"?

Theorem ($|\mathbb{R}|$ (Cantor 1877))

$$|(0,1)| = |\mathbb{R}| = |\mathbb{R} \times \mathbb{R}| = |\mathbb{R}^n|$$

"Je le vois, mais je ne le crois pas !"

"I see it, but I don't believe it !"

— Cantor's letter to Dedekind (1877).

Q : Then, what is "dimension"?

Theorem (Brouwer (Topological Invariance of Dimension))

There is no continuous bijections between $\mathbb{R}^m$ and $\mathbb{R}^n$ for $m \neq n$.

Beyond

C

Theorem (Cantor's Theorem (1891))

$$|A| \neq |\mathcal{P}(A)|$$

Theorem (Cantor's Theorem (1891))

$$|A| \neq |\mathcal{P}(A)|$$

Theorem (Cantor Theorem)

If $f : A \to \mathcal{P}(A)$, then f is not onto.

Theorem (Cantor Theorem)

$$|A| < |\mathcal{P}(A)|$$

Theorem (Cantor Theorem)

$$|A| < |\mathcal{P}(A)|$$

$$A \qquad \mathcal{P}(A) \qquad \mathcal{P}(\mathcal{P}(A)) \qquad \ldots$$

Theorem (Cantor Theorem)

$$|A| < |\mathcal{P}(A)|$$

$$A \qquad \mathcal{P}(A) \qquad \mathcal{P}(\mathcal{P}(A)) \qquad \ldots$$

There is no largest infinity.

$$<$$

Definition ($|A| \leq |B|$)

$|A| \leq |B|$ if there exists an *one-to-one* function f from A into B.

Definition ($|A| < |B|$)

$$|A| < |B| \iff |A| \leq |B| \wedge |A| \neq |B|$$

Definition ($|A| < |B|$)

$$|A| < |B| \iff |A| \leq |B| \wedge |A| \neq |B|$$

$$|\mathbb{N}| < |\mathbb{R}|$$

$$|X| < |2^X|$$

$$|\mathbb{N}| < |2^{\mathbb{N}}|$$

Definition (Countable Revisited)

X is countable:

$$(\exists n \in \mathbb{N} : |X| = n) \lor |X| = |\mathbb{N}|$$

Definition (Countable Revisited)

X is countable:

$$(\exists n \in \mathbb{N} : |X| = n) \lor |X| = |\mathbb{N}|$$

Theorem (Proof for Countable)

X is countable iff

$$|X| \leq |\mathbb{N}|.$$

Definition (Countable Revisited)

X is countable:

$$(\exists n \in \mathbb{N} : |X| = n) \vee |X| = |\mathbb{N}|$$

Theorem (Proof for Countable)

X is countable iff

$$|X| \leq |\mathbb{N}|.$$

X is countable iff there exists a one-to-one function

$$f : X \to \mathbb{N}.$$

Subsets of Countable Set

Every subset B of a countable set A is countable.

Subsets of Countable Set

Every subset B of a countable set A is countable.

$$\exists f : A \xrightarrow{1-1} \mathbb{N}$$

Subsets of Countable Set

Every subset B of a countable set A is countable.

$$\exists f : A \xrightarrow{1-1} \mathbb{N}$$

$$g = f|_B$$

Slope

(a) The set of all lines with rational slopes

Slope

(a) The set of all lines with rational slopes

$$(\mathbb{Q},\ \mathbb{R})$$

Slope

(a) The set of all lines with rational slopes

$$(\mathbb{Q},\ \mathbb{R})$$

$$|\mathbb{R}| \leq\ |\mathbb{Q} \times \mathbb{R}|\ \leq |\mathbb{R} \times \mathbb{R}| = |\mathbb{R}|$$

Q : *Is "$\leq$" a partial order?*

Q : Is "$\leq$" a partial order?

Theorem (Cantor-Schröder–Bernstein (1887))

$$|X| \leq |Y| \wedge |Y| \leq |X| \implies |X| = |Y|$$

Q : *Is "$\leq$" a partial order?*

Theorem (Cantor-Schröder–Bernstein (1887))

$$|X| \leq |Y| \wedge |Y| \leq |X| \implies |X| = |Y|$$

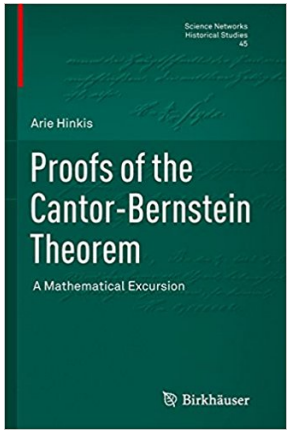
$$\exists\, f : X \xrightarrow{1-1} Y \wedge g : Y \xrightarrow{1-1} X \implies \exists\, h : X \xleftrightarrow[onto]{1-1} Y$$

Q : *Is "$\leq$" a partial order?*

Theorem (Cantor-Schröder–Bernstein (1887))

$$|X| \leq |Y| \wedge |Y| \leq |X| \implies |X| = |Y|$$

$$\exists\, f : X \xrightarrow{1-1} Y \wedge g : Y \xrightarrow{1-1} X \implies \exists\, h : X \xleftrightarrow[onto]{1-1} Y$$

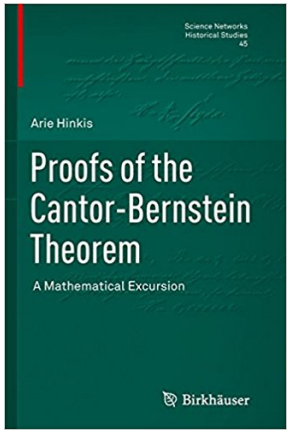


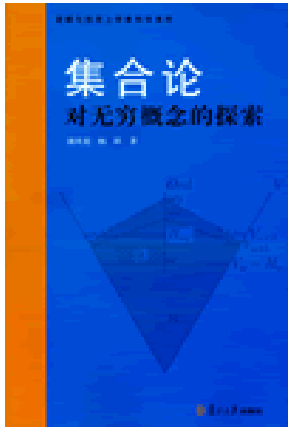
Q : *Is "$\leq$" a partial order?*

Theorem (Cantor-Schröder–Bernstein (1887))

$$|X| \leq |Y| \wedge |Y| \leq |X| \implies |X| = |Y|$$

$$\exists\, f : X \xrightarrow{1-1} Y \wedge g : Y \xrightarrow{1-1} X \implies \exists\, h : X \xleftrightarrow[onto]{1-1} Y$$



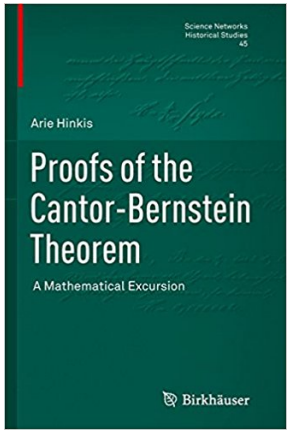


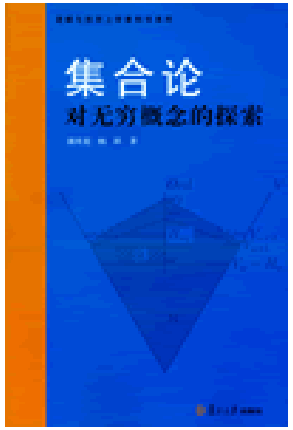
Q : *Is "$\leq$" a partial order?*

Theorem (Cantor-Schröder–Bernstein (1887))

$$|X| \leq |Y| \wedge |Y| \leq |X| \implies |X| = |Y|$$

$$\exists\, f : X \xrightarrow{1-1} Y \wedge g : Y \xrightarrow{1-1} X \implies \exists\, h : X \xleftrightarrow[onto]{1-1} Y$$





Schröder–Bernstein theorem @ wiki

Q : *Is "$\leq$" a total order?*

Q : *Is "$\leq$" a total order?*

Theorem (PCC)

Principle of Cardinal Comparability (PCC) $\iff$ *Axiom of Choice*

Theorem

$$|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = |\mathcal{P}(\mathbb{Q})|$$

Theorem

$$|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = |\mathcal{P}(\mathbb{Q})|$$

$$|\mathbb{R}| \leq |\mathcal{P}(\mathbb{Q})| \qquad |\mathcal{P}(\mathbb{Q})| \leq |\mathbb{R}|$$

Theorem

$$|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = |\mathcal{P}(\mathbb{Q})|$$

$$|\mathbb{R}| \leq |\mathcal{P}(\mathbb{Q})| \qquad |\mathcal{P}(\mathbb{Q})| \leq |\mathbb{R}|$$

https://en.wikipedia.org/wiki/Cardinality_of_the_continuum#Cardinal_equalities

Theorem

$$|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = |\mathcal{P}(\mathbb{Q})|$$

$$|\mathbb{R}| \leq |\mathcal{P}(\mathbb{Q})| \qquad |\mathcal{P}(\mathbb{Q})| \leq |\mathbb{R}|$$

https://en.wikipedia.org/wiki/Cardinality_of_the_continuum#Cardinal_equalities

$$\mathfrak{c} \triangleq |\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = |2^{\mathbb{N}}| \triangleq 2^{\aleph_0}$$

Theorem

$$|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = |\mathcal{P}(\mathbb{Q})|$$

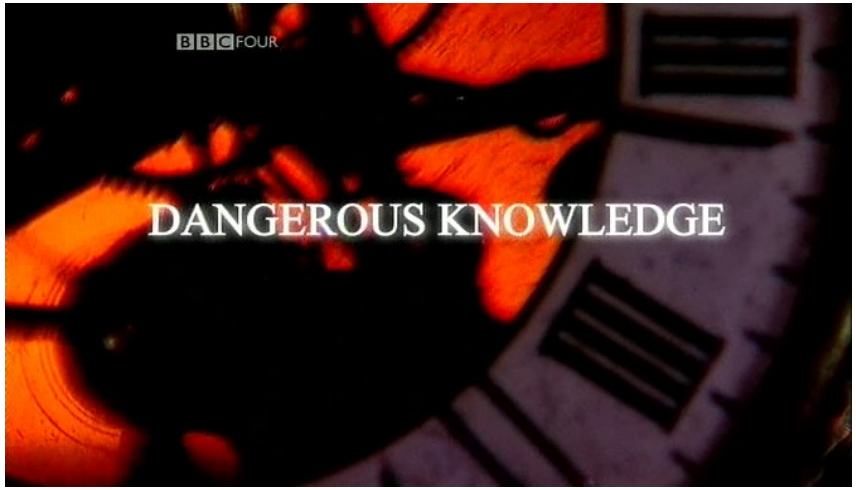
$$|\mathbb{R}| \leq |\mathcal{P}(\mathbb{Q})| \qquad |\mathcal{P}(\mathbb{Q})| \leq |\mathbb{R}|$$

https://en.wikipedia.org/wiki/Cardinality_of_the_continuum#Cardinal_equalities

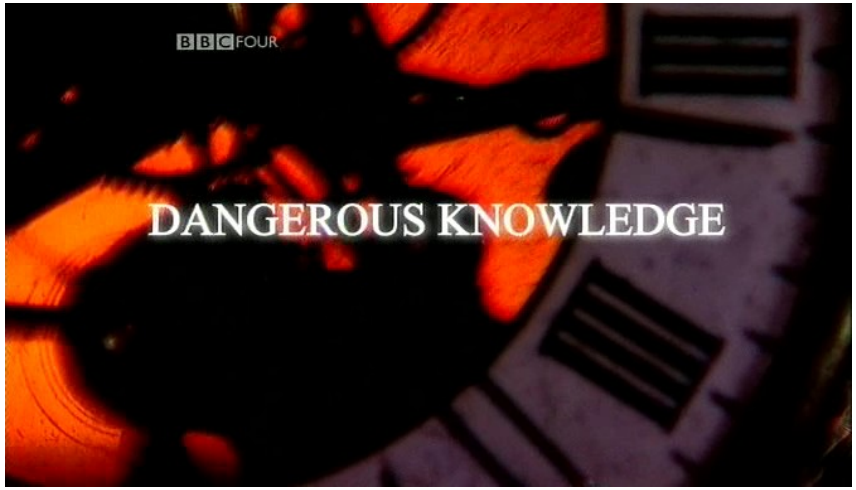
$$\mathfrak{c} \triangleq |\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = |2^{\mathbb{N}}| \triangleq 2^{\aleph_0}$$

Continuum Hypothesis (CH)

$$\boxed{\nexists A : \aleph_0 < |A| < \mathfrak{c}}$$



☞ Dangerous Knowledge (22:20; BBC 2007)



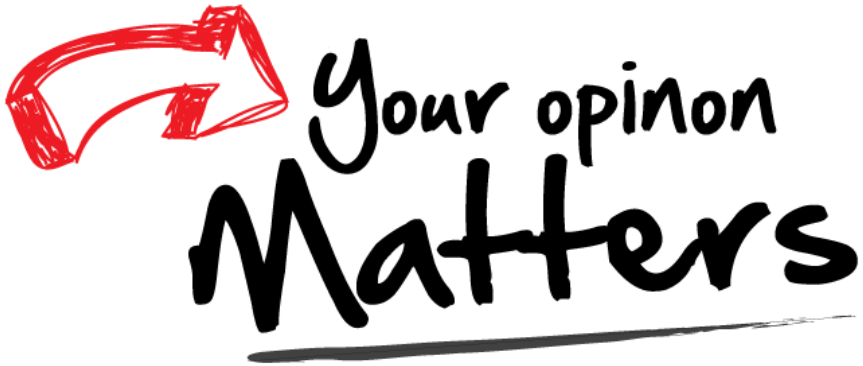
☞ Dangerous Knowledge (22:20; BBC 2007)

Independence from ZFC:

Kurt Gödel (1940) CH cannot be disproved from ZF.

Paul Cohen (1964) CH cannot be proven from the ZFC axioms.

Thank You!



hfwei@hnu.edu.cn